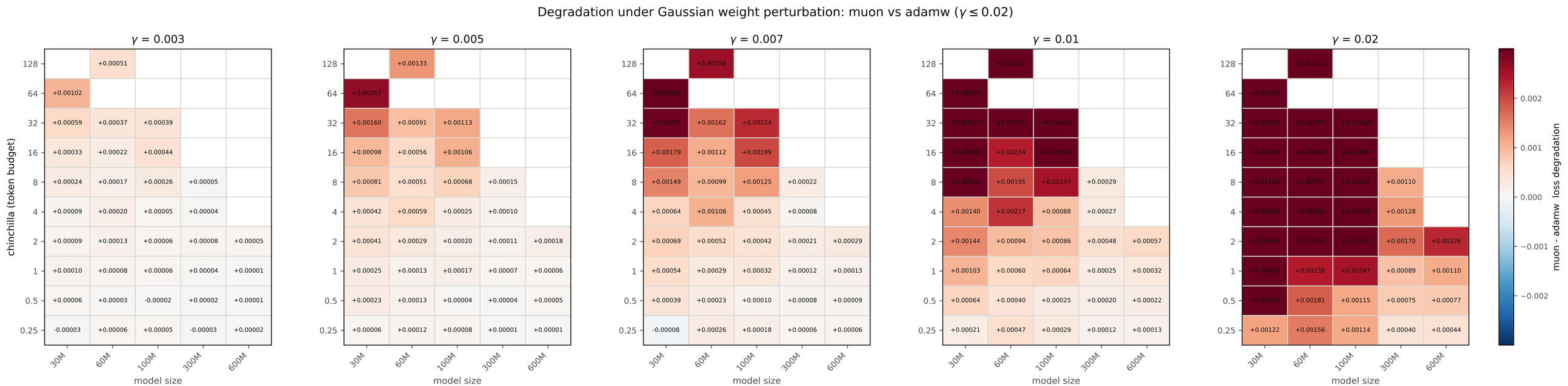
Degradation under Gaussian weight perturbation: muon vs adamw ($\gamma \leq 0.02$)
γ = 0.003
γ = 0.005
γ = 0.007
γ = 0.01
γ = 0.02
chinchilla (token budget)
128 64 32 16 8 4 2 1 0.5 0.25
30M 60M 100M 300M 600M
model size
+0.00051 +0.00102 +0.00059 +0.00037 +0.00039 +0.00033 +0.00022 +0.00044 +0.00024 +0.00017 +0.00026 +0.00005 +0.00009 +0.00020 +0.00005 +0.00004 +0.00009 +0.00013 +0.00006 +0.00008 +0.00005 +0.00010 +0.00008 +0.00006 +0.00004 +0.00001 +0.00006 +0.00003 -0.00002 +0.00002 +0.00001 -0.00003 +0.00006 +0.00005 -0.00003 +0.00002
+0.00133 +0.00267 +0.00160 +0.00091 +0.00113 +0.00098 +0.00056 +0.00106 +0.00081 +0.00051 +0.00068 +0.00015 +0.00042 +0.00059 +0.00025 +0.00010 +0.00041 +0.00029 +0.00020 +0.00011 +0.00018 +0.00025 +0.00013 +0.00017 +0.00007 +0.00006 +0.00023 +0.00013 +0.00004 +0.00004 +0.00005 +0.00006 +0.00012 +0.00008 +0.00001 +0.00001
+0.00258 +0.00491 +0.00291 +0.00162 +0.00224 +0.00179 +0.00112 +0.00199 +0.00149 +0.00099 +0.00125 +0.00022 +0.00064 +0.00108 +0.00045 +0.00008 +0.00069 +0.00052 +0.00042 +0.00021 +0.00029 +0.00054 +0.00029 +0.00032 +0.00012 +0.00013 +0.00039 +0.00023 +0.00010 +0.00008 +0.00009 -0.00008 +0.00026 +0.00018 +0.00006 +0.00006
+0.00521 +0.00909 +0.00577 +0.00327 +0.00443 +0.00363 +0.00234 +0.00378 +0.00300 +0.00195 +0.00247 +0.00029 +0.00140 +0.00217 +0.00088 +0.00027 +0.00144 +0.00094 +0.00086 +0.00048 +0.00057 +0.00103 +0.00060 +0.00064 +0.00025 +0.00032 +0.00064 +0.00040 +0.00025 +0.00020 +0.00022 +0.00021 +0.00047 +0.00029 +0.00012 +0.00013
+0.02111 +0.03799 +0.02328 +0.01270 +0.01826 +0.01496 +0.00944 +0.01485 +0.01192 +0.00781 +0.00963 +0.00110 +0.00583 +0.00843 +0.00366 +0.00128 +0.00585 +0.00354 +0.00340 +0.00170 +0.00226 +0.00430 +0.00238 +0.00247 +0.00089 +0.00110 +0.00302 +0.00181 +0.00115 +0.00075 +0.00077 +0.00122 +0.00156 +0.00114 +0.00040 +0.00044
muon - adamw loss degradation
0.002 0.001 0.000 −0.001 −0.002
cell = (perturbed - unperturbed) for muon minus the same for adamw | negative (blue) = muon degrades less | white = no data
colour scale clipped at ±0.003 (85th pct); 27 cell(s) beyond it keep their printed value